# ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** six | Two months each |
| 2 | **(b)** troposphere | Weather layer |
| 3 | **(a)** rise | Low pressure |
| 4 | **(c)** from sea to land | Land warmer |
| 5 | **(d)** sink | High pressure |
| 6 | **(a)** monsoon | Heavy rain |
| 7 | **(a)** greenhouse gases released by activities | Emission measure |
| 8 | **(c)** June to September | Rainy season |
| 9 | **(b)** mesosphere | Friction |
| 10 | **(c)** December and January | Winter |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Envelope of gases around the Earth (1); Held by gravity (1). | 1+1 |
| 12 | Greenhouse gases released by activity (1); Measured as total emissions (1). | 1+1 |
| 13 | Seasonal reversal of winds (1); Brings rain to India (1). | 1+1 |
| 14 | Water falling from clouds (1); Rain, snow, hail (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Up to 80 km (1); Meteors burn (1); Temperature falls with height (1). | 1+1+1 |
| 16 | Sea breeze (1); Land breeze (1); Both are daily (1). | 1+1+1 |
| 17 | Up to 50 km (1); Cloud-free (1); Ozone layer (1). | 1+1+1 |
| | **OR:** Average 12 km (1); Weather happens (1); Temperature falls with height (1). | 1+1+1 |
| 18 | Farming (1); Water supply (1); Economy and festivals (1). | 1+1+1 |
| | **OR:** Temperature and humidity (1); Precipitation (1); Pressure and wind (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Weather: today's rain (1); Climate: 30-year pattern (1); Weather changes fast (1); Climate stable (1); India has tropical monsoon climate (1). | 1+1+1+1+1 |
| | **OR:** Winter (1); Summer (1); Monsoon (1); Post-monsoon (1); Months and features (1). | 1+1+1+1+1 |
| 20 | Troposphere 12 km with weather (1); Stratosphere 50 km with ozone (1); Temperature falls then rises (1); Aeroplanes fly in stratosphere (1); Boundaries tropopause and stratopause (1). | 1+1+1+1+1 |
| | **OR:** Light near the poles (1); Solar wind (1); Borealis (1); Australis (1); Thermosphere (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# THE MASTERMIND COACHING CLASSES

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** atmosphere | Blanket of air |
| 2 | **(b)** carbon dioxide | Heat trapping |
| 3 | **(d)** human activities | Fossil fuels |
| 4 | **(c)** season | Arabic |
| 5 | **(c)** stratosphere | Protects from UV |
| 6 | **(a)** exosphere | Very thin |
| 7 | **(a)** precipitation | Falling moisture |
| 8 | **(b)** west | Named from direction |
| 9 | **(c)** nitrogen | 78% |
| 10 | **(a)** 30 years or more | Long term |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Nitrogen (1); 78% (1). | 1+1 |
| 12 | Day-to-day state of the atmosphere (1); Changes quickly (1). | 1+1 |
| 13 | Stratosphere (1); Absorbs ultraviolet rays (1). | 1+1 |
| 14 | Moving air (1); From high to low pressure (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Fossil fuels (1); Deforestation (1); Pollution (1). | 1+1+1 |
| 16 | Rain and drizzle (1); Snow and sleet (1); Hail (1). | 1+1+1 |
| 17 | Heavy rain (1); Weak embankments (1); Building near rivers (1). | 1+1+1 |
| | **OR:** Rises rapidly in temperature (1); Reflects radio waves (1); Auroras occur (1). | 1+1+1 |
| 18 | Weather short-term, climate long-term (1); Weather changes quickly (1); Climate average (1). | 1+1+1 |
| | **OR:** Troposphere, stratosphere (1); Mesosphere (1); Thermosphere, exosphere (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Greenhouse gases trap heat (1); Fossil fuels and deforestation (1); Effects on floods, droughts (1); Use renewable energy (1); Plant trees and save water (1). | 1+1+1+1+1 |
| | **OR:** Falls in troposphere (1); Rises in stratosphere (1); Falls in mesosphere (1); Rises in thermosphere (1); Exosphere very thin (1). | 1+1+1+1+1 |
| 20 | Troposphere (1); Stratosphere (1); Mesosphere (1); Thermosphere (1); Exosphere (1). | 1+1+1+1+1 |
| | **OR:** Agriculture (1); Water (1); Economy (1); Festivals (1); Floods or drought risk (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

## ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** south-west monsoon | Summer monsoon |
| 2 | **(a)** oxygen | 21% |
| 3 | **(a)** a short time | Day-to-day |
| 4 | **(a)** summer | April-June |
| 5 | **(c)** saves energy | Reduces footprint |
| 6 | **(b)** troposphere | Clouds and rain |
| 7 | **(b)** reduce carbon footprint | Absorbs CO2 |
| 8 | **(a)** high pressure to low pressure | Pressure difference |
| 9 | **(d)** Weather Ready and Climate Smart | Forecasting |
| 10 | **(d)** km/hr | Speed |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Water vapour in the air (1); From evaporation (1). | 1+1 |
| 12 | Long-term change in weather pattern (1); Caused by humans (1). | 1+1 |
| 13 | Average weather over a long period (1); 30 years or more (1). | 1+1 |
| 14 | Troposphere (1); Contains water vapour (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | South-west monsoon (1); North-east monsoon (1); Different seasons (1). | 1+1+1 |
| 16 | Nitrogen 78% (1); Oxygen 21% (1); Argon 0.93% (1). | 1+1+1 |
| 17 | Floods and droughts (1); Melting glaciers (1); Rising sea level (1). | 1+1+1 |
| | **OR:** Walk or cycle (1); Save water (1); Reuse and recycle (1). | 1+1+1 |
| 18 | Crops damaged (1); Houses destroyed (1); People moved to relief camps (1). | 1+1+1 |
| | **OR:** Winter and summer (1); Monsoon (1); Post-monsoon (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Day: sea breeze (1); Night: land breeze (1); Differential heating (1); Pressure difference (1); Moderates coastal climate (1). | 1+1+1+1+1 |
| | **OR:** Vasanta spring (1); Grishma summer (1); Varsha monsoon (1); Sharad and Hemanta (1); Shishira winter (1). | 1+1+1+1+1 |
| 20 | Nitrogen 78% (1); Oxygen 21% (1); Argon and CO2 (1); Water vapour (1); Dust (1). | 1+1+1+1+1 |
| | **OR:** Rivers Satluj, Beas, Ravi (1); Heavy monsoon rain (1); Weak embankments (1); Crop and house damage (1); Lessons for flood management (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.